

Priyanka Chatterjee

AI-Enabled Software Engineer · Microservices & Event-Driven Architecture ·
AWS/Azure/GCP · 9,300 ★ OSS Contributor · 3.5+ Years
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SUMMARY

Software Engineer with 3.5+ years of experience and expertise in **AI-powered software development**, **scalable microservices**, **event-driven architecture**, and cloud platforms (**AWS, Azure, GCP**). Built a Model Context Protocol server and multi-agent LLM automation for enterprise **compliance** workflows at Turbot HQ, cutting resource-discovery latency 70-90% while meeting CIS, NIST, and SOC 2 regulatory standards.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, JavaScript (ES6+), SQL, Bash

AI Engineering: LLM Application Development, Agentic Workflows, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), OpenAI Agents SDK, Pydantic AI

Software Architecture: Microservices, Event-Driven Architecture, Distributed Systems, Serverless, REST, gRPC, OAuth 2.0 (Auth0), JWT

Cloud Platforms: AWS (Lambda, EC2, S3, SecurityHub, GuardDuty), Azure, GCP

Messaging & Streaming: Apache Kafka (consumer groups, DLQs), AWS SNS/SQS, AWS EventBridge, Azure Event Grid

Data Privacy, Security & Compliance: CIS (AWS, Azure, GCP, M365, GitHub, Google Workspace), NIST, HIPAA, PCI DSS v4, SOC 2, FedRAMP; IAM, RBAC, cloud security policies

Core CS: Data Structures & Algorithms, Object-Oriented Design, Low-Level & High-Level System Design, API Design

Databases: PostgreSQL, MySQL, DuckDB, MongoDB, Redis (ElastiCache)

DevOps & Testing: Docker, GitHub Actions, Git, Unit Testing, Integration Testing, Code Review

WORK EXPERIENCE

Turbot HQ, Inc. Remote

August 2022 – April 2026

Software Engineer I

- Designed a multi-agent **AI architecture** on the OpenAI Agents SDK, translating a business **security** requirement (preventing public-cloud misconfigurations) into an automated technical solution: LLM agents that generate policy commands, validate them in a live test environment, self-correct failures, and score accounts against an internal **security** metric.
- Built the Steampipe Model Context Protocol (MCP) server end to end in Python, embedding **AI** capabilities directly into a production enterprise platform to give LLM agents standardized, real-time access to cloud and **security** data.
- Designed and developed an internal REST API microservice in Go using the Gin framework as part of a **scalable microservices architecture** powering Turbot Guardrails, a **security-compliance** SaaS platform; secured endpoints with OAuth 2.0 authentication via Auth0, containerized with Docker, deployed to **AWS** EC2, and served as a gRPC client for several internal **microservices**.
- Built a Kafka-based, **event-driven** pipeline where consumers asynchronously processed policy-enforcement events and fired a webhook once a policy was successfully enforced against a resource; added a dead-letter queue (DLQ) for reliability.
- Built a RAG-based Pydantic **AI** agent deployed as a **serverless AWS** Lambda function to resolve cross-team business queries, ingesting internal documentation into S3 vector storage and surfacing it through Slack.
- Authored multi-cloud **compliance** modules covering 1,000+ resource checks across CIS, NIST, HIPAA, PCI DSS v4, SOC 2, and FedRAMP regulatory frameworks for enterprise customers in finance, healthcare, and government — cutting audit time 50%.
- Mentored teammates on adopting **AI**-assisted engineering tools (Claude Code, Cursor) into their daily workflows.

PROJECTS

General-Purpose AI Chatbot / Agent (Model Context Protocol) (Jun 2025 – Aug 2025)

Agentic chatbot/agent framework built with Pydantic **AI** featuring configurable external MCP servers via a persistent MongoDB-backed registry, JSON config loader with strict input validation, per-server failure isolation, native tool calling, mid-conversation multi-model switching (OpenAI, Gemini, Anthropic, Mistral), conversation-history persistence, sync + streaming output, prompt-injection safeguards, and Logfire observability.

EDUCATION

University of Engineering & Management (UEM) Kolkata, WB, IN

August 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.28

Relevant Coursework: Object-Oriented Programming, DBMS, Data Structures and Algorithms, Shell Scripting in Linux, Low-Level Design, High-Level Design